TOR DIMENSION AT MOST D IS EQUIVALENT TO THE EXISTENCE OF A FLAT RESOLUTION OF LENGTH D

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- Characterisation of tor amplitude by complexes of flat modules. Lemma \ref{lemma-tor-amplitude}, cited at proof lines 17014-17015. Turns a flat resolution of length d into the bound on tor dimension.
- The truncation of a projective resolution in degrees at least minus d is a flat resolution of length d. Lemma \ref{lemma-last-one-flat}, cited at proof lines 17019-17020. Gives the converse direction.

Source reference. Original lemma 066F: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 7934–7934. The lemma's own canonical tag, quoted so that readers can resolve the label inside the delivered text; it points at the counted statement itself and carries no proof content.

Source reference. Original lemma 0654: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 7888–7888. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 0653: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 7887–7887. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Lemma 0.2. *Let R be a ring. Let M be an R-module. Let $d \geq 0$. The following are equivalent*

(1) *M has tor dimension $\leq d$, and*

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(2) *there exists a resolution*

$$0 \to F_d \to \ldots \to F_1 \to F_0 \to M \to 0$$

with F_i a flat R-module.

In particular an R-module has tor dimension 0 if and only if it is a flat R-module.

Proof. Assume (2). Then the complex $E^\bullet$ with $E^{-i} = F_i$ is quasi-isomorphic to M. Hence the Tor dimension of M is at most d by Lemma 0654. Conversely, assume (1). Let $P^\bullet \to M$ be a projective resolution of M. By Lemma 0653 we see that $\tau_{\geq -d} P^\bullet$ is a flat resolution of M of length d, i.e., (2) holds. □